

Modeling Substructures in Embedded Disks: Evidence for Early Planet Formation

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Introduction

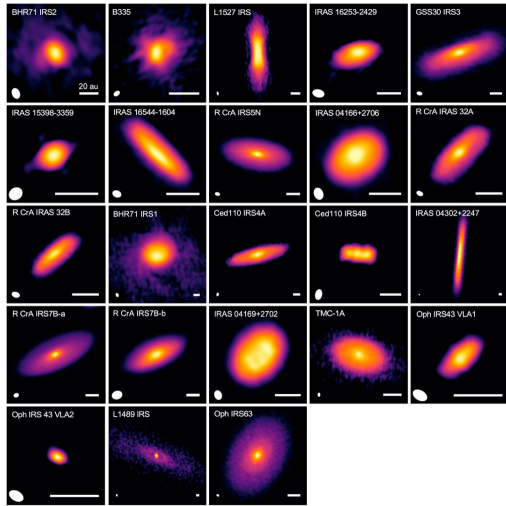


Figure 1. ALMA 1.3 mm continuum images of 24 embedded disks from the eDisk program. Each panel shows one source, with scale bar showing 20 au.

Embedded disks (eDisk) are the earliest forms of protoplanetary disks, still deeply enshrouded in the gas and dust cloud from which they formed. Our work involves visualizing the content of these disks using visibilities collected from the **Atacama Large Millimeter/submillimeter Array (ALMA)**. We want to investigate substructures as evidences that early planet formation is already underway in these embedded disks.

Methods

The visualization process involves running different **2D analytic models** to fit different sources through **MCMC** process. We use a residuals plot, the model data subtracted from the raw data, in order to depict hidden structures in the protoplanetary disk, like a **ring**, **gap**, or **spiral**. These hidden substructures reveal more about the formation process of protoplanets. Our main model used is **exponentially tapered power-law model**, which has a higher central concentration and steeper drop at larger radii. Our work involves developing new models based on careful analysis of residuals and radial profiles, and enhancing the model fitting codebase, such as allowing simultaneous fitting on binary disks.

Results

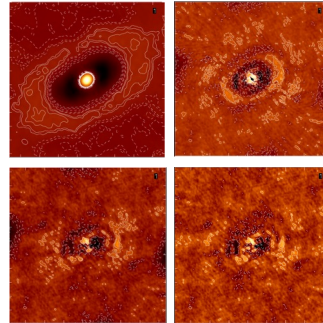


Figure 2. Residuals of Oph IRS63 with model Gaussian, exponentially tapered power-law (2-gamma), power law + gap + central Gaussian, and exponentially tapered broken weighted power-law model, which is the current bestfit model. Oph IRS63 is a source with two known rings.

*Note that the residuals have flipped and rotated orientation compared with the source image

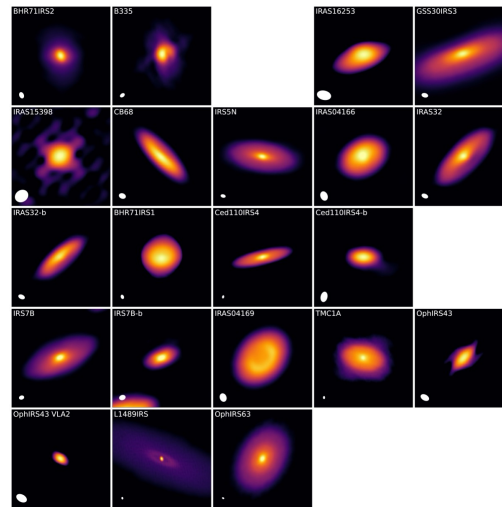


Figure 3. Current best fit model gallery. The missing two sources are edge-on sources, which were modeled separately.

Most of the sources can be well fitted by an exponentially tapered power-law model with an asymmetric profile. For several more complicated sources, best fit models are specifically designed through radial profile and residuals analysis. Extra structures including asymmetric sin ring, spiral arms, broken power law, asymmetric radius, and Gaussian envelope with different inclination are involved in these best fit models.

Discussion & Conclusion

Our modeling for the eDisk sources is largely complete, and we are now in the process of carefully interpreting the results. In particular, we are analyzing the brightness profiles of our best-fit models to evaluate the physical plausibility of the inferred substructures. It is essential to distinguish between features that reflect real structures in the source and those that arise from optical depth effects or model fitting artifacts without clear physical interpretation.

A comprehensive paper summarizing our modeling results is underway and will be published in *The Astrophysical Journal*.

Reference

Ohashi, N., Tobin, J. J., Jørgensen, J. K., Takakuwa, S., Sheehan, P., Aikawa, Y., Li, Z.-Y., & Looney, L. W. (2023). Early Planet Formation in embedded disks (eDisk). I. Overview of the program and first results. *The Astrophysical Journal*, 951(1), 8. <https://doi.org/10.3847/1538-4357/acd384>

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